

Information Retrieval using Semantic Search

PROGRAM:

```
dictionary = {
    "the", "longest", "list", "of", "stuff",
    "at", "domain", "name", "long", "last"
}

def word_segment(s, dictionary):
    n = len(s)

    dp = [False] * (n + 1)
    dp[0] = True

    words_used = [""] * (n + 1)

    for i in range(1, n + 1):
        for j in range(i):
            if dp[j] and s[j:i] in dictionary:
                dp[i] = True
                words_used[i] = s[j:i]
                break

    if not dp[n]:
        return None

    result = []
    i = n
    while i > 0:
        word = words_used[i]
        result.insert(0, word)
        i -= len(word)

    return result

text = "thelongestlistofthelongeststuffatthelongestdomainnameatlonglast"

result = word_segment(text, dictionary)
if result:
    print("Segmented Words:", result)
else:
    print("No valid segmentation found")
```

OUTPUT:

```
IDLE Shell 3.13.12
File Edit Shell Debug Options Window Help
Python 3.13.12 (tags/v3.13.12:1cbe481, Feb  3 2026, 18:22:25) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\kames\OneDrive\Documents\semantic.py =====
Segmented Words: ['the', 'longest', 'list', 'of', 'the', 'longest', 'stuff', 'at', 'long', 'last']
>>> |
```